

In ASP.NET (Core or MVC), every incoming HTTP call is represented by an `HttpContext` object. Within it you'll find two key properties:

- **`HttpContext.Request`** – everything the client (browser, mobile app, API consumer) sends to your server.
- **`HttpContext.Response`** – everything your application sends back to the client.

HttpContext.Request (Incoming HTTP Request)

Represents the full HTTP request message. Key parts include:

Element	What it Represents	Examples
Method	HTTP verb	GET, POST, PUT, DELETE
Scheme	Protocol	http or https
Host / Path / QueryString	Target URL pieces	Host: example.com, Path: /products/42, QueryString: ?sort=price
Headers	Key-value pairs carrying metadata	User-Agent, Accept, Authorization, Cookie
Body	The request payload (for POST/PUT)	JSON { "name": "Widget" }, form fields, file uploads
Cookies	Parsed cookies sent by the client	<code>Request.Cookies["sessionId"]</code>
Content-Type & Length	MIME type and size	application/json, text/html
Protocol / HTTP Version	Version of HTTP	HTTP/1.1, HTTP/2
Client Info	Connection details	Remote IP, ports

These components together represent the **Request Line + Headers + Optional Body** defined by the HTTP specification.

HttpContext.Response (Outgoing HTTP Response)

Represents the full HTTP response your app returns. Key parts include:

Element	What it Represents	Examples
Status Code	Result of the request	200 OK, 404 Not Found, 500 Internal Server Error
Headers	Metadata about the response	Content-Type, Set-Cookie, Cache-Control
Body	The payload sent to the client	HTML, JSON, file bytes
Cookies	Cookies to set or update	Response.Cookies.Append("sessionId", "...")
Content-Type & Length	MIME type and size	application/json, image/png
Redirects	Location header for 3xx responses	Response.Redirect("/login")

Together these correspond to the Status Line + Headers + Optional Body of an HTTP response.